Programming Worlds

PROGRAMMING WORLDS ACADEMY

Git & GitHub 101

From Zero to Your First Pull Request

BEYOND CODE. BEYOND LIMITS.

09

MODULE 09

Collaboration and Pull Requests

Propose changes on a branch, get them reviewed, and merge them on GitHub.

MODULE 09 · LESSON 9.1

The branch-and-push workflow

سير العمل بالفروع

Learning objectives

- Describe the team workflow.
- Create and push a feature branch.
- Know collaborators versus forks.

Prerequisites

Modules 7 and 8.

Introduction

Teams never commit straight to main. They work on branches and propose changes.

Explanation

Workflow. Update main, create a branch, commit, push the branch, open a pull request, get a review, merge, then clean up (Figure 1).

Start every task with `git switch main` and `git pull` so your branch starts from the latest work.

Create the branch: `git switch -c feature/about-page`. Use short, clear names: `feature/…`, `fix/…`, `docs/…`.

Commit small, focused changes with clear messages.

Publish: `git push -u origin feature/about-page`. GitHub then shows a yellow banner with **Compare & pull request**.

Collaborator. The owner invites you under *Settings → Collaborators*. You can push branches to the same repository.

Fork. If you are not a collaborator, you click **Fork** to make your own copy, push there, and open a pull request to the original. Forks are how open-source contributions work.

Real-world analogy

> You write your chapter in a separate document and hand it to the editor. The editor decides when it joins the book.

Diagram

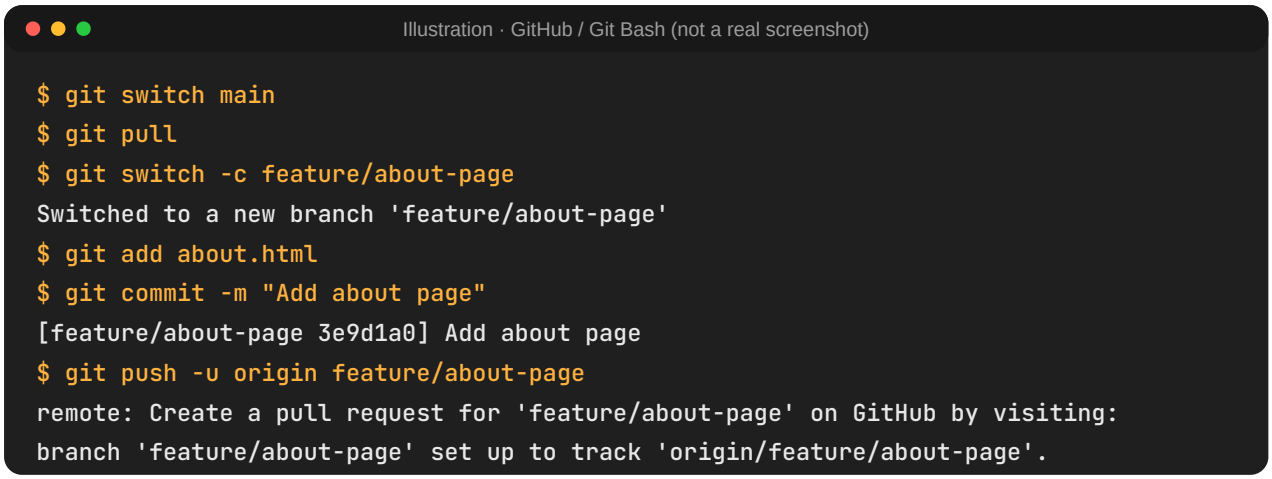


Figure 1 · Illustration: branch, commit, push

Commands

`git switch main`

`git pull`

`git switch -c feature/about-page`

`git push -u origin feature/about-page`

Figure 1 shows these commands in order.

Common mistakes and solutions

Branching from an old main.	Pull main first.
Huge branches with many unrelated changes.	One purpose per branch.
Pushing to main on a protected repository.	Push a branch instead.

Practical exercise

> Create `feature/about-page`, add `about.html`, commit, and push it.

Expected result

GitHub lists the branch and offers Compare & pull request.

Knowledge check

1. Where do teams commit new work?

- ○ Nowhere
- ○ On a branch
- ○ Directly on main

2. What does -u do in the push?

- ○ Sets the upstream
- ○ Deletes
- ○ Uploads twice

3. When do you fork?

- ○ Never
- ○ You cannot push to the original
- ○ Always

Answers

- On a branch — Branches protect main.
- Sets the upstream — Later you can just git push.
- You cannot push to the original — Contributors use forks.

Summary

Pull main, branch, commit, push the branch. Forks are for outside contributors.

Key vocabulary

Feature branch	Branch for one change
Fork	Your own GitHub copy of someone else's repository
Collaborator	Person with write access

Next lesson

Lesson 9.2 — Opening a pull request.

MODULE 09 · LESSON 9.2

Opening a pull request

فتح طلب دمج

Learning objectives

- Define a pull request.
- Open one on GitHub.
- Write a good title and description.

Prerequisites

Lesson 9.1.

Introduction

A pull request (PR) asks the team to review and merge your branch into another branch.

Explanation

On GitHub, click **Compare & pull request** (or the **Pull requests** tab → **New pull request**).

Check the two branches at the top: **base** is where changes go (main) and **compare** is your branch.

Write a clear **title**, for example `Add about page`.

Write the **description**: what changed, why, and how to check it. Mention an issue with `Closes #3` if there is one.

Look at the **Files changed** tab. Review your own diff before asking others.

Click **Create pull request**. Optionally choose reviewers on the right. Use **Create draft pull request** if the work is not ready.

Any new commit you push to the same branch appears in the PR automatically.

Real-world analogy

> A PR is a cover letter attached to your chapter: it explains what you changed and invites feedback.

Diagram

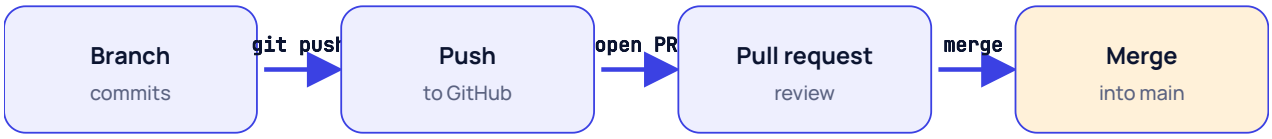


Figure 1 · The pull request flow

Commands

No new Git command is needed. A PR lives on GitHub; you push commits as usual.

Figure 1 summarises the pull request flow.

Common mistakes and solutions

Base set to the wrong branch.	Check base and compare before creating.
Empty description.	Always explain what and why.
Opening a PR with unrelated changes.	Split into separate PRs.

Practical exercise

> Open a pull request from `feature/about-page` into main with a title and description.

Expected result

The PR page shows your commit and “Able to merge”.

Knowledge check

1. What is a pull request?

- ○ A Git command
- ○ A request to review and merge a branch
- ○ A download

2. Which is the base branch?

- ○ Where changes will go
- ○ Your branch
- ○ The remote

3. A draft PR means…

- ○ Deleted
- ○ Not ready for review
- ○ Already merged

Answers

- A request to review and merge a branch — PRs live on GitHub.
- Where changes will go — Usually main.
- Not ready for review — It signals work in progress.

Summary

A PR proposes a branch for review. Give it a clear title and description.

Key vocabulary

Pull request	Proposal to merge a branch
base	Target branch
compare	Your branch

Next lesson

Lesson 9.3 — Review and merge.

MODULE 09 · LESSON 9.3

Review, merge and clean up

المراجعة والدمج والتنظيف

Learning objectives

- Review a PR.
- Merge on GitHub.
- Update local main and delete the branch.

Prerequisites

Lesson 9.2.

Introduction

A review makes the code better, and merging makes it official.

Explanation

Reviewers open **Files changed**, hover a line, click the blue **+**, and write a comment. **Review changes** lets them **Comment**, **Approve** or **Request changes**.

To answer feedback, commit to the same branch and push. The PR updates itself.

When approved and checks pass, click **Merge pull request** then **Confirm merge**. Other options are **Squash and merge** (one commit) and **Rebase and merge**. This course uses the default merge.

GitHub shows **Delete branch**. Click it; the work is safe in main.

Locally: `git switch main`, `git pull` (Figure 1), then `git branch -d feature/about-page`.

If the PR shows conflicts, pull main into your branch (`git pull origin main` while on your branch), resolve them as in Lesson 7.3, and push.

Be kind in reviews: comment on the work, suggest, and explain.

Real-world analogy

> The editor reads your chapter, leaves notes, and finally approves it into the book. You then update your own copy of the book.

Diagram



Figure 1 · Illustration: cleaning up after a merge

Commands

`git switch main`
`git pull`
`git branch -d feature/about-page`

Optional: `git push origin --delete feature/about-page` removes the remote branch if GitHub did not.

Common mistakes and solutions

Forgetting to pull after merging on GitHub.	Your local main is still old until you pull.
Merging your own PR without review in a team.	Follow team rules.
Deleting a branch before it is merged.	Merge first.

Practical exercise

> Merge your PR, delete the branch on GitHub, pull main locally, and delete the local branch.

Expected result

about.html exists on main locally and `git branch` shows only main.

Knowledge check

1. Who clicks Merge pull request?

- ○ Git automatically
- ○ Someone with permission after approval
- ○ Anyone anonymously

2. After merging on GitHub you should…

- ○ Pull main locally
- ○ Reinstall
- ○ Fork again

3. Request changes means…

- ○ The repo is deleted
- ○ Fixes are needed before merge
- ○ The PR is merged

Answers

- Someone with permission after approval — Teams define the rules.
- Pull main locally — Sync your local copy.
- Fixes are needed before merge — Update and push.

Summary

Review, approve, merge, delete the branch, then pull main.

Key vocabulary

Approve	Reviewer accepts the PR
Squash and merge	Merge as a single commit
Request changes	Reviewer asks for fixes

Next lesson

> **Module 10 — Undoing mistakes.**

MODULE 09

Practical lab 9 · Your first pull request

1. `git switch main` and `git pull`.
2. `git switch -c feature/about-page`.
3. Create `about.html` with a heading and a paragraph. Commit: `Add about page`.
4. `git push -u origin feature/about-page`.
5. On GitHub, click **Compare & pull request**, write a title and description, and create the PR.
6. Review **Files changed**, then **Merge pull request**, then **Delete branch**.
7. Locally: `git switch main`, `git pull`, `git branch -d feature/about-page`.

Check	Pass if…
PR	PR page exists with a description
Merge	about.html is on main on GitHub
Cleanup	Only main remains locally

Evaluation: 3 of 3 = pass.

Module 9 quiz (pass mark: 70%)

1. Where do you do new work?

- ○ On main
- ○ Offline only
- ○ On a branch

2. What is a PR?

- ○ A clone
- ○ A request to review and merge
- ○ A download

3. What do you do after merging on GitHub?

- ○ Pull main locally
- ○ Fork
- ○ Nothing

4. What does Approve mean?

- ○ Push failed
- ○ Reviewer accepts the changes
- ○ Branch deleted

5. Why write a description?

- ○ It is required by Git
- ○ For fun
- ○ So reviewers understand what and why

Answers

- On a branch — Branches protect main.
- A request to review and merge — It lives on GitHub.
- Pull main locally — Sync your computer.
- Reviewer accepts the changes — It allows merge.
- So reviewers understand what and why — Context speeds reviews.